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2.3.5 SVD of nonsingular matrices

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singular values, maybe there's relationship there. Hmm, we said that the matrix starts behaving badly if it's almost singular. I wonder what we can say about the matrix being almost singular from the singular values. Hmm, had something to do with the condition number of the matrix. I wonder how we might be able to express the inverse of matrix A in terms of its singular value decomposition? I wonder what the

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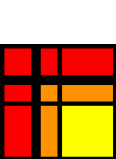
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Reading assignment

1/1 point (graded)



Read Unit 2.3.5 of the notes. [\[LINK\]](#)

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💬 [Ponder This 2.3.5.6.](#)

I think the image of the unit ball would look like a highly eccentric ellipse as...

1

Homework 2.3.5.1.

1/1 point (graded)
Let $\mathbf{A} \in \mathbb{C}^{m \times m}$ and $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H$ be its SVD.

$\mathbf{A}$ is nonsingular if and only if $\mathbf{\Sigma}$ is nonsingular.

TRUE

 Answer: TRUE

For full solution see notes: [[LINK](#)]

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Homework 2.3.5.2.

1/1 point (graded)
Let $\mathbf{A} \in \mathbb{C}^{m \times m}$ and $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H$ be its SVD with

$$\mathbf{\Sigma} = \begin{pmatrix} \sigma_0 & 0 & \cdots & 0 \\ 0 & \sigma_1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_{m-1} \end{pmatrix}$$

$\mathbf{A}$ is nonsingular if and only if $\sigma_{m-1} \neq 0$.

TRUE

 Answer: TRUE

For full solution see notes: [[LINK](#)]

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Homework 2.3.5.3.

1/1 point (graded)
Let $\mathbf{A} \in \mathbb{C}^{m \times m}$ be nonsingular and $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H$ be its SVD.

The SVD of $\mathbf{A}^{-1}$ equals $\mathbf{V}\mathbf{\Sigma}^{-1}\mathbf{U}^H$.

SOMETIMES

 Answer: SOMETIMES

For full solution see notes: [[LINK](#)]

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Homework 2.3.5.4.

1/1 point (graded)
Let $\mathbf{A} \in \mathbb{C}^{m \times m}$ be nonsingular and

$$\begin{aligned}
 \mathbf{A} &= \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H \\
 &= \left(\begin{array}{c|c|c} \mathbf{u}_0 & \cdots & \mathbf{u}_{n-1} \end{array} \right) \left(\begin{array}{c|c|c} \hline \sigma_0 & \cdots & 0 \\ \hline \vdots & \ddots & \vdots \\ \hline 0 & \cdots & \sigma_{m-1} \\ \hline \end{array} \right) \left(\begin{array}{c|c|c} \mathbf{v}_0 & \cdots & \mathbf{v}_m \end{array} \right)
 \end{aligned}$$

be its SVD.

The SVD of $\mathbf{A}^{-1}$ is given by (indicate all correct answers):

☐
 $\mathbf{V}\mathbf{\Sigma}^{-1}\mathbf{U}^H$

☐
 $\left(\begin{array}{c|c|c} \mathbf{v}_0 & \cdots & \mathbf{v}_{m-1} \end{array} \right) \left(\begin{array}{c|c|c} \hline 1/\sigma_0 & \cdots & 0 \\ \hline \vdots & \ddots & \vdots \\ \hline 0 & \cdots & 1/\sigma_{m-1} \\ \hline \end{array} \right) \left(\begin{array}{c} \mathbf{u}_0 \\ \vdots \\ \mathbf{u}_{m-1} \end{array} \right)$

☒
 $\left(\begin{array}{c|c|c} \mathbf{v}_{m-1} & \cdots & \mathbf{v}_0 \end{array} \right) \left(\begin{array}{c|c|c} \hline 1/\sigma_{m-1} & \cdots & 0 \\ \hline \vdots & \ddots & \vdots \\ \hline 0 & \cdots & 1/\sigma_0 \\ \hline \end{array} \right) \left(\begin{array}{c} \mathbf{u}_m \\ \vdots \\ \mathbf{u}_0 \end{array} \right)$

☒
 $(\mathbf{V}\mathbf{P}^H)(\mathbf{P}\mathbf{\Sigma}^{-1}\mathbf{P}^H)(\mathbf{U}\mathbf{P}^H)^H$ where $\mathbf{P} = \begin{pmatrix} 0 & \cdots & 0 \\ 0 & \cdots & 1 \\ \vdots & & \vdots \\ 1 & \cdots & 0 \end{pmatrix}$



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Homework 2.3.5.5.

1/1 point (graded)
 Let $\mathbf{A} \in \mathbb{C}^{m \times m}$ be nonsingular.

$$\|\mathbf{A}^{-1}\|_2 = 1/\min_{\|x\|_2=1} \|\mathbf{A}x\|_2.$$

ALWAYS

 Answer: ALWAYS

For full solution see notes: [[LINK](#)]

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